
Below the Noise Floor: A Deterministic GLEE Agent and the Measurement Problem It Exposed

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GLEE Agent ID edb1db91303e (7aidara_Gamma)

Agent track: 6th place. Human track: 2nd place.

Abstract

I competed in both GLEE tracks: as 7AIDARA_GAMMA, a deterministic, LLM-free agent that finished **6th in the agent track** (2360.8) over **264,432 games and 3.11M logged turns**, and by hand, where I finished **2nd in the human track**. The agent is $\approx 2,300$ lines of Python, one closed-form policy per family behind 114 parameters, with no learning and no language model. My central finding is methodological and negative: **GLEE’s percentile-Elo cannot resolve the effects entrants are trying to produce**. Five *byte-identical* builds, polled simultaneously for 8.6 hours over $\approx 3,000$ games each, differed by up to 88 rating points overall and 293 within a family, while their *behaviour* differed by ≤ 0.010 of the pot; separating two policies that differ by 0.25 rating points per game at 3σ needs $\approx 5,600$ – $6,700$ games *per arm*. Across 30+ interventions and 49 runs only *strictly dominated* fixes ever replicated, every tuning arm measured nothing, and one unmeasured arm shipped fleet-wide cost 100–385 negotiation points per agent. Much of what the policies rely on is not documented anywhere and had to be recovered from play: the valuation pool, an undisclosed round cap that pays both sides nothing, and the rating’s own update rule. I give four offline techniques that measured what the leaderboard could not, a survivorship failure mode that silently corrupted four analyses, and a measurement of what abandoning games actually costs. Playing both tracks yields an unusually controlled human-versus-machine comparison, one designer and one shared opponent pool, which finds that agent and human read seller messages equally well (hit rate 0.738 vs. 0.725) and separate almost entirely on the discipline to decline.

1 Introduction

GLEE [1], in the lineage of automated negotiation competitions [9, 6], evaluates agents in three economic families: *bargaining* (alternating-offer division with discounting), *negotiation* (bilateral trade under private valuations [5]) and *persuasion* (a 20-round sender–receiver game). Configurations come from a grid of 960 combinations, and payoff is scored not in dollars but as a *percentile* against every payoff earned on the same configuration in the same role, adjusted for opponent strength: $r_{\text{game}} = 2000 + 8000(\text{pct} - 0.5)$, $\Delta R = \eta(r_{\text{game}} - R)$, $\eta \rightarrow 0.002$.

Two consequences shaped every decision. A \$0 no-deal sits at the bottom of that scale, so closing a mediocre deal strictly dominates holding out and missing. And the step size I recovered from my own rating deltas, $\eta = 0.002$, implies an EMA half-life of ≈ 346 games *per family*: the rating is a smoothed, lagged, opponent-contaminated statistic, and it is the only feedback the platform gives.

I built a deterministic agent rather than an LLM one for three reasons. Percentile scoring over a known parametric family rewards a correct decision rule; the free-text channel measured as inert (§5); and determinism buys what mattered most, namely that **a logged game replays to exactly the**

same decisions, which made offline evaluation possible once I found that the leaderboard could not evaluate anything.

One theme runs through what follows: **most of what the agent exploits is undocumented and had to be inferred from play**, so every such fact is stated with the evidence that recovered it. The paper also spans the workshop’s three levels: what the agent did (§6), what I did facing the same games by hand (§7), and where the two diverge. I contribute the agent (§2), a measurement of the competition’s evaluation noise floor and the sample it implies (§3), four offline techniques that measure strategy changes without the leaderboard (§4), a ledger of what replicated and what cost me places (§5), the required behaviour analysis (§6), and a same-author human baseline (§7).

2 The agent, and what had to be discovered to build it

A dispatcher routes each game by family to one of three closed-form policies¹ (858 / 660 / 781 lines), set by 114 tunables and covered by 472 behavioural tests. Any policy that raises is replaced by a legal fallback: abandonment scores at the **5th percentile** and three self-timeouts trigger a 30-minute queue ban, so a crashing policy is far worse than a mediocre one. No game in 264,432 was lost to an exception.

Bargaining anchors on the infinite-horizon fixed point $(1 - \delta_{\text{them}}) / (1 - \delta_{\text{us}} \delta_{\text{them}})$ [2] rather than the finite recursion, which swings between ≈ 0.24 and ≈ 0.91 with the horizon’s *parity* and is right only against an opponent playing the same equilibrium: the agent banks 0.364 where it says 0.236. Parity still buys a *guarantee*, since if the last proposal is mine the agent can refuse everything and reach it, priced at what that seat empirically returns (91.9% signed, $n=1,415$). It never walks away, since that pays what running out of rounds pays.

Negotiation rests on a fact the documentation does not state. Sorting every valuation the server dealt me showed they are not continuous but drawn from three fixed four-rung pools, (80, 100, 120, 150) scaled by $\{1, 10^2, 10^4\}$, both players on the same scale. My own value therefore pins the whole ladder, so the agent prices at *tradable rungs* walked from the top down instead of guessing a reservation price. Roles are drawn independently of values, so only 37.5% of hidden-value games hold any surplus and a seat with none is walked at once.

Persuasion rests on a second undocumented fact, verified over 1,829 sold rounds: the seller is paid the full price on *every* sale, high or low, so reputation is the only currency. Bayesian persuasion [3] then fixes the tolerable lie rate: recommending every high unit and a fraction q of low ones puts the buyer’s posterior at $p / (p + (1-p)q)$, so $q^* = p(1-\tau) / (\tau(1-p))$ for threshold τ . The agent spends under it, ramping as rounds run out (Appendix Table A1). A third undocumented fact, found from a cluster of \$0 outcomes at high round numbers, reshaped both open-horizon policies: games sold as unlimited stop at a round cap paying *both* sides nothing.

3 The measurement problem

Because only an account’s *best* agent is ranked, my five slots could be spent as five simultaneous *controls* rather than five experiments (Appendix Table A2), which turns an unfalsifiable rating into a measured null. Polled together for 8.6 hours over $\approx 3,000$ games each, five **byte-identical** builds spread a mean of 48.1 rating points and up to 292.9 within a family, while their *behaviour* differed by ≤ 0.010 of the pot (Appendix Figure A1).

The size of that noise is arithmetic. The rating is an 80-fold magnification of a percentile, so with η decayed the displayed rating inverts to $\text{pct} = 0.5 + (R - 2000) / 8000$, evaluated where the field’s payoff density is highest. A 346-point bargaining collapse was a move from the 51.8th percentile to the 47.4th, and the final board is compressed the same way: **ranks 3 to 7 of the agent track are separated by 0.41 percentile points**, and the 10.5 rating points by which I missed 5th are 0.13 of one. Appendix D has three further properties of the rule.

Two further properties compound it. The pool is **non-stationary**, since scoring is against the research dataset *plus competition games as they are played*, so *no cross-time comparison is valid*; and

¹**LLM usage.** The agent contains no language model. I used an LLM coding assistant for analysis scripts and drafting under my direction, and verified every reported number against the logs.

ratings **freeze when idle**, making the close a stopping problem. I banked +83 points by freezing an agent on a one-hour high, and on the final evening three of the four accounts above me had already stopped playing. Worse, **almost every run is underpowered**: the per-game rating delta has $sd=4.84/4.42/4.44$ over 120,023 games, so separating two arms differing by Δ at 3σ needs $n=(3\sigma\sqrt{2}/\Delta)^2$ games *each*, or **6,743 / 5,632 / 5,689 per arm at $\Delta=0.25$** , against effects of 0.05–0.20 and a best window of $\approx 4,000$ games per agent.

4 Measuring without the leaderboard

Four offline techniques carried essentially all my evidence, using only my own logs; the fourth, which recovers the quality rate on *declined* rounds by subtraction from the known prior, is in Appendix B.

(A) Paired within-game counterfactuals. Accepting is *unilateral and terminal*, so the other branch is arithmetic: signing now is exactly $\text{share} \times \delta^{(r-1)}$, which makes every refused offer its own control, with no survivorship. It gave a break-even accept bar per configuration from 23,617 refusals.

(B) Decision-level replay, as a mandatory pre-flight. The turn logs re-run the shipped decision function exactly, reproducing **4,272 of 4,272** bargaining and **13,056 of 13,056** negotiation decisions with zero mismatches, so the replay *is* the agent. Count the decisions a flag *changes*, which is *not* how often its condition fires: one arm’s condition held on 43% of eligible games and changed **0 of 13,056** decisions, another **0 of 13,479**.

(C) Break-even recovery. ΔR is monotone in percentile, so the outcome at which mean ΔR crosses zero is the outcome ranking at my own standing, the field median when $R \approx 2000$. Fitting ΔR on banked share within a configuration and taking the root uses *every* game, \$0 no-deals included, so there is no survivorship (slope positive in 58 of 64 cells, median $R^2=0.60$, split-half $r=0.741$). It localised my bargaining deficit to a single seat, and showed the loss there was the accept bar [7], not the opening ask. Across all 55 cells it also characterises the field: its median tracks Rubinstein SPE where that prediction is interior ($r=0.895$, mean gap 0.070 of pot) and abandons it where it is extreme (gap 0.281), taking **0.297** where theory says take nothing and **0.734** where it says take everything [10]. An equilibrium anchor is therefore miscalibrated at the grid’s edges (Appendix Tables A3 and A4).

The failure mode these replace. Four analyses were wrong the same way: **the filter that built the sample already excluded every game that would have contradicted it** (Appendix C).

5 What worked, what failed, and what it cost

Six changes replicated, each removing a *strictly dominated* choice, an option plainly worse than an available alternative (Appendix Table A5).

Nothing else replicated. Seven consecutive tuning arms landed inside the noise floor, as did a seller lie-rate arm held at 100% against 40% controls across three runs ($t=-0.29$). **The most expensive mistake was mine**: on the final morning I shipped one negotiation concession parameter to *all four* unfrozen agents at once and left it five hours before measuring. It cost **100–385 negotiation points per agent**, and I finished 10.5 points behind 5th; one retained control would have caught it in an hour.

Abandoning games costs less than it appears to. An abandoned game scores at the 5th percentile, and stopping a run kills what is in flight: I abandoned **80 of 109,881** rated games (0.073%), 78 of them inside seven clock hours, at -6.76 rating points against $+0.17$ for a game played out. The steps sum to -541 , but the rating is an EMA, so the persistent shift is about **0.5 points**; the close is the exception, where nothing is averaged away.

My two best changes came from reading transcripts, not aggregates. One is a parser bug: purchases made against messages saying, in plain English, “*Not this one... pass this round.*” The refusal list held “not this round” but not “not this one”, and such near-misses fell through a default branch that read anything unmatched as a recommendation. Re-scoring 123,700 buyer rounds, 270 of the re-classified rounds were purchases and **258 of them were low quality**. The other was a 96-round stall showing that my acceptance gate opened at round 97 in games the server killed at 96. A third reading settled the architecture: **language was inert**, the free-text channel moving buyer take-up only from 0.689 to 0.693 [4].

6 Agent Behavior Analysis

What the agent actually did. Bargaining settled at median round 2 with a 1.5% no-deal rate, banking 0.429 of the pot. Negotiation correctly walked 79.8% of untradable seats, and agreed 20–24% under hidden values against 85–89% under complete information. The persuasion buyer shows the intended discontinuity: 17.4 purchases per game where buying blind pays, 2.2 where it does not, and the patient-hold gate fires on **60.7%** of player-1 $\delta=0.95$ decisions against **0.0%** of player-2 $\delta=1.0$.

How alignment is enforced. The 472 tests assert the decision the policy must make in a named regime rather than a utility: at HEAD all pass, while the final night’s working tree, still carrying that run’s arm flags, fails 27, each naming the behaviour its flag changed. Randomisation is seeded from (game_id, round), so the agent is a pure function of observed state, and technique (B) reproduces 100% of 17,328 logged decision points.

Four cases where it did not match design. (1) **Rules that could never fire:** three flags changed zero decisions or were unreachable, so intent and behaviour were different things. (2) **A safety valve keyed to the wrong horizon:** my accept collapse armed two rounds from a cap recorded as 99, in a family whose games died at 96, so it never fired and 78.8% of open negotiations ended at \$0. (3) **A parser whose default was the wrong way round,** which made the failure silent. (4) **A schedule that met its target and lost anyway:** 86% of games ground to the last round and dumped for *any* positive profit at -6.4 rating per game, which deal rate could not see because the capitulation *is* a deal (Appendix Table A6).

7 A same-author human baseline

Table 1: Human versus agent on identical configurations. Bargaining: mean banked share, known horizon. Persuasion: buyer seat, by whether buying *blind* pays.

Bargaining (known horizon)				Persuasion (buyer seat)			
seat	cells	human	agent	regime	human	agent	<i>t</i>
player 1 (first mover)	14	0.444	0.447	$mp > 1$ (blind gain)	+24.22	+22.10	+1.31
player 2 (last word)	13	0.512	0.720	$mp = 1$ (knife edge)	+2.99	+2.56	+0.56
				$mp < 1$ (blind loss)	−0.61	+0.18	−2.44

I also played the *human* track by hand and finished **2nd** over 2,690 rated games against the same pool. Reconstructed from per-game deltas my ratings were 2437/2406/2170 against the agent’s 2128/2818/2136; since the tracks rank separately I rest the analysis on *matched configurations* (Table 1).

(1) **Agent and human read sellers equally well:** hit rate on purchases 0.738 over 2,745 for me, 0.725 over 59,603 for the agent, so the intuition that a person reads deception better than a keyword list is unsupported. (2) **The machine’s edge is patience, and the seat it unlocks:** holding the final proposal it banks 0.720 of the pot against my 0.512, and refusing was correct in **96.5%** of its 13,180 measurable games against **70.4%** of my 455. In the first-mover seat the two are level. (3) **My cost is discipline, not perception:** where participation does not pay I still buy 8.4 units per game in cells the agent declines ($t = -2.44$), and banking zero costs only -0.04 to -3.15 points where a *negative* payoff costs -4.96 to -7.65 .

8 Limitations, reproducibility, and conclusion

Every number here is confounded by a non-stationary pool and by one account in one competition; technique (C) assumes ΔR is locally linear; and the human sample is one person. The largest gap is **cross-game memory** [8]: honesty profiles are stable (split-half $r = 0.786$) but exploiting them needs state the architecture lacks (Appendix B). Much of the edge came from undocumented facts recovered from my own logs, a reproducibility hazard as much as an advantage; the agent itself is reproducible, being three modules, one parameter file, 472 offline tests, and every turn logged. Four lessons remain: **measure your noise floor first; prefer strictly dominated fixes; replay before you run; read transcripts.** A leaderboard is a scoreboard, not an instrument.

References

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A Supplementary figure and tables

Table A1: Seller-script ladder (persuasion buyer seat): forward hit rate on the *next* purchase, conditioned on what that exact message template has already delivered earlier in the same game (6,185 buyer replays). The 5.3% row sits below the break-even bar $1/m$ in every cell, which is what makes the template veto strictly dominated.

Prior record of this template	hit rate	Prior record	hit rate
no prior	70.4%	all-low $\times 1$	33.8%
all-high $\times 1$	81.3%	all-low $\times 2$	19.3%
all-high $\times 2$	84.8%	all-low $\times 3$	5.3%
all-high $\times 3$	91.2%	mixed	63.7%

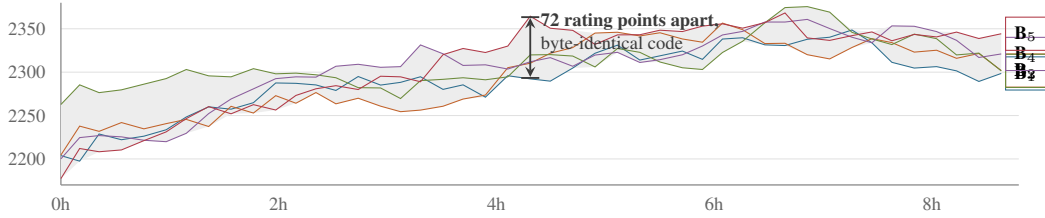


Figure A1: **Five byte-identical builds, 8.6 h, 45 simultaneous polls, $\approx 3,000$ games each.** Band = min–max envelope per poll (mean spread 48.1, max 88.0). Per family the same builds spread 136.8 / 161.8 / 109.8 points on average and 292.9 at peak, while their *behaviour* differed by ≤ 0.010 of the pot.

Table A2: Fleet volume, 12–30 August 2026, recovered from the local turn logs. Only the account’s best slot is ranked, which is what makes the other four available as simultaneous controls.

Family	unique games	Fleet total	
Bargaining	89,641	logged turns	3,114,356
Negotiation	76,490	unique games	264,432
Persuasion	98,301	reconstructable runs	49

Table A3: Break-even recovery (technique C) over 7,104 known-horizon and 31,208 open-horizon bargaining games. The whole bargaining deficit is one cell, and it is the accept side: replaying the shipped decision rule over 26,773 logged decision points (98.79% control fidelity) shows the stonewall term setting the bar as low as 0.118, so it signs 0.30–0.50 offers in the opening rounds.

Horizon	Seat	I bank	Field median	Gap	Diagnosis
open	p_1	0.530	0.476	+0.054	healthy
open	p_2	0.501	0.463	+0.038	healthy
known 12	p_1	0.435	0.816	−0.381	accept bar collapses to 0.118
known 12	p_2	0.687	0.449	+0.239	healthy

Table A4: Open-horizon bargaining: the field median is approximately Rubinstein SPE, and I sit under it in every player-1 cell (technique C, 38,312 games). The exception is $\delta_{us}=\delta_{them}=0.95$, where SPE is 0.513 but the field median is only 0.475, so demanding SPE there buys no-deals, which is why the shipped floor targets the measured field median rather than the equilibrium.

δ_{us}	δ_{them}	mine	field median	SPE	gap
0.95	0.80	0.712	0.825	0.833	-0.11
0.95	0.90	0.536	0.625	0.690	-0.09
0.90	0.80	0.595	0.675	0.714	-0.08
0.90	0.90	0.449	0.525	0.526	-0.08
1.00	0.90	0.692	0.775	1.000	-0.08
0.80	0.80	0.477	0.525	0.556	-0.05

Table A5: Five of the six changes that replicated. Each removed a *strictly dominated* choice, an option plainly worse than an available alternative with no trade-off to price. Nothing else did.

Change	Evidence
Complete-info ultimatum	I was asking 0.650 of a <i>known</i> surplus on the one round they cannot refuse; 0.99 was signable. $+3.42\sigma$, $n=214$
Rung-aware pricing	Price at the recovered valuation rungs, not guesses. $+3.9/+4.1\sigma$, replicated
Free-clock accept floor	At $\delta=1$ refusing is costless. $+0.0458$ of pot, $n=108$, 7.7σ , none worse
Open-horizon cap collapse	The server zeroes both sides and never says so. Caught 11 of 11 \$0 games
Negative-signal parser	270 buys made against messages that read as refusals; 95.6% low quality against a 27.5% base rate

Table A6: Bounded complete-information negotiation: the same seat under two horizons, over 10,740 games. Grinding to the last round is where the family’s losses are, and deal rate cannot see it, because the round-10 dump *is* a deal.

Cell / outcome	n	banked / buyer value	rating / game
max_rounds=1, seller, complete info	488	asks 0.991, signed 99.2%	+6.10
max_rounds=10, seller, complete info	469	(split below)	-4.63
closed by round 8	66	0.2365	+5.9
ground to round 10	403	0.0433	-6.4

B Extended limitations

Three limitations were compressed out of §7. (i) **The persuasion buyer seat admits no per-round counterfactual even in principle:** quality on declined rounds is never revealed, so technique (D) answers only the aggregate question “am I leaking in this cell”, never “should I have bought round 7”. (ii) **The offline simulator is optimistic.** I built a local bargaining engine (real rules, my seat played by the shipped code, the other seat by a field fitted to 15,788 real games plus eight adversarial archetypes). It reproduces per-cell median settlement rounds but scores $\approx +0.03$ of pot high, with a ± 45 -rating noise floor per world at $n=4,000$, so I used it only to choose which arm deserved a live window and never shipped on it alone. (iii) **Operational failure modes cost me more than any strategy error.** Three recur: a deploy that writes arm flags back into the repository working tree (hit five times, and the reason 27 tests fail on my final-night tree); an entry point outside the copied package, so sibling agents silently ran stale code; and force-killing an agent mid-game, which abandons in-flight games at the 5th percentile and can trip a 30-minute queue ban. Draining with a deadline rather than a kill is the only safe stop.

(iv) **Opponent modelling is stable but unreachable where it would pay.** §8 reports split-half $r=0.786$ across the 19 opponent $\times$ cell profiles with ≥ 25 purchases in both halves, which is real but is also the whole usable sample. Three things block the rest. Identity is withheld in 49.9% of games; the named half spreads 16,498 games over 227 opponents and 15 cells, about five games per profile, and behaviour differs by cell so they cannot be pooled; and quality is redacted on every

round I passed (0 of 42,616 unbought rounds ever showed it), so a reputation table has *zero* entries in precisely the cells where I buy nothing and most need one. That is why the shipped buyer reads the seller’s rationing rate within the current game, which is visible whether or not I buy, instead of a cross-game profile.

C A note on the Simpson’s paradox I nearly shipped on

A late candidate rule, “take a recommendation only from a seller who has already spent a sale to decline one”, looked refuted when pooled across all fifteen (m, p) cells: subsequent dud rates of 0.285 / 0.287 / 0.286 / 0.315 for sellers who had declined 0 / 1 / 2 / 3–4 of their first four rounds. Split by cell it reverses and becomes monotone: inside $(m=1.20, p=0.80)$, $n=9,932$, the same buckets read 0.160 / 0.102 / 0.081 / **0.069** against a break-even of $1 - 1/m = 0.167$. The pooled number is dominated by easy cells where everyone buys everything regardless. I did not ship it, for the reason the rest of this paper argues: at 2.2% of my games it was worth $\approx +2.6$ rating points, far below what the remaining window could measure, and a fourth simultaneous change would have destroyed my ability to read the three already running.

D Three properties of the scoring rule

The aggregate rewards balance, and I optimised the wrong object for two weeks. Overall rank is the *unweighted mean* of the three family ratings, and only an account’s best slot is ranked. Judged per family, negotiation was my best return per game; judged per *overall* point it was nearly worthless, because +300 more negotiation is +100 overall and hard, while +180 in each weak family is +120 and was what the arms already targeted. At the time I noticed, I sat 11th at 2301.5 with negotiation 2875 (4th best on the board) and bargaining 2021; the 4th-placed account held that same bargaining rating. **Nobody in the top ten was balanced**, which is the strongest structural incentive the scoring rule creates and the one I was slowest to see.

The reference pool defends itself against sabotage. A natural worry is that spare agent slots could be spent depressing rivals’ ratings. Besides being prohibited and trivially detectable, it is self-defeating on the mechanism: percentile is scored against payoffs earned on the same configuration, and competition games join that pool as they are played, so every no-deal forced on an opponent injects a bottom-of-distribution payoff that *raises* the percentile of every normal payoff scored afterwards. Attacking the field inflates it. There is also no opponent selection anywhere in the API, so the damage would spray uniformly across the 227 opponents I faced.

The underlying data will come from the organisers rather than from me. Every number here is computed from my own turn logs, which also carry the transcripts of the agents I was matched against, and those are not released alongside the code. The organisers have said they will publish the full game record for every agent, which supersedes anything I could have attached: it covers the whole field rather than one account, and it comes from the party that ran the games. The analysis scripts are released now, so the method can be audited ahead of the data it will run on.

The garden of forking paths is wide here. One persuasion arm, analysed three defensible ways on the same data, read -3.11σ across 3 cells, -0.82σ across 33, and $+3.02\sigma$ across 44, depending only on the inclusion threshold for how many games a configuration needed to enter the sample. Pre-registering the cell filter before looking is not optional at this noise level.